



Melody

Breaking the Algorithmic Echo Chamber

An AI-Powered Music Discovery Assistant



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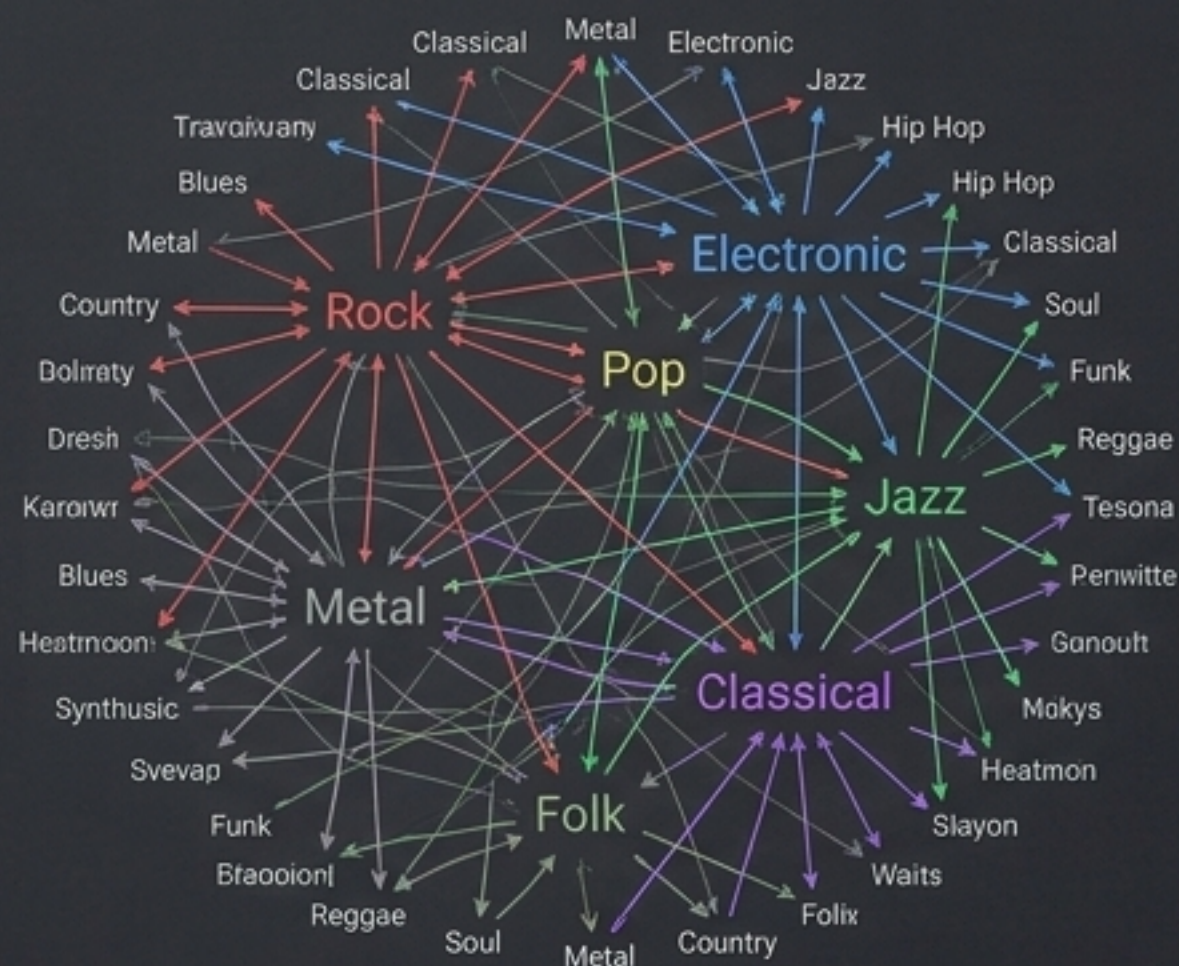
Education

Bachelor's degree student in Computer Science at the Open University

Quote

'I'm a die-hard tech geek and a musician. I love everything that bridges these domains: from coding, FPV drones, and ESP32 hardware, to playing instruments and exploring the endless world of music.'

THE ECHO CHAMBER



Standard music recommendation systems trap listeners in repetitive loops, returning only shallow similarity matches and preventing true discovery.

DEEP DISCOVERY



TARGET USER

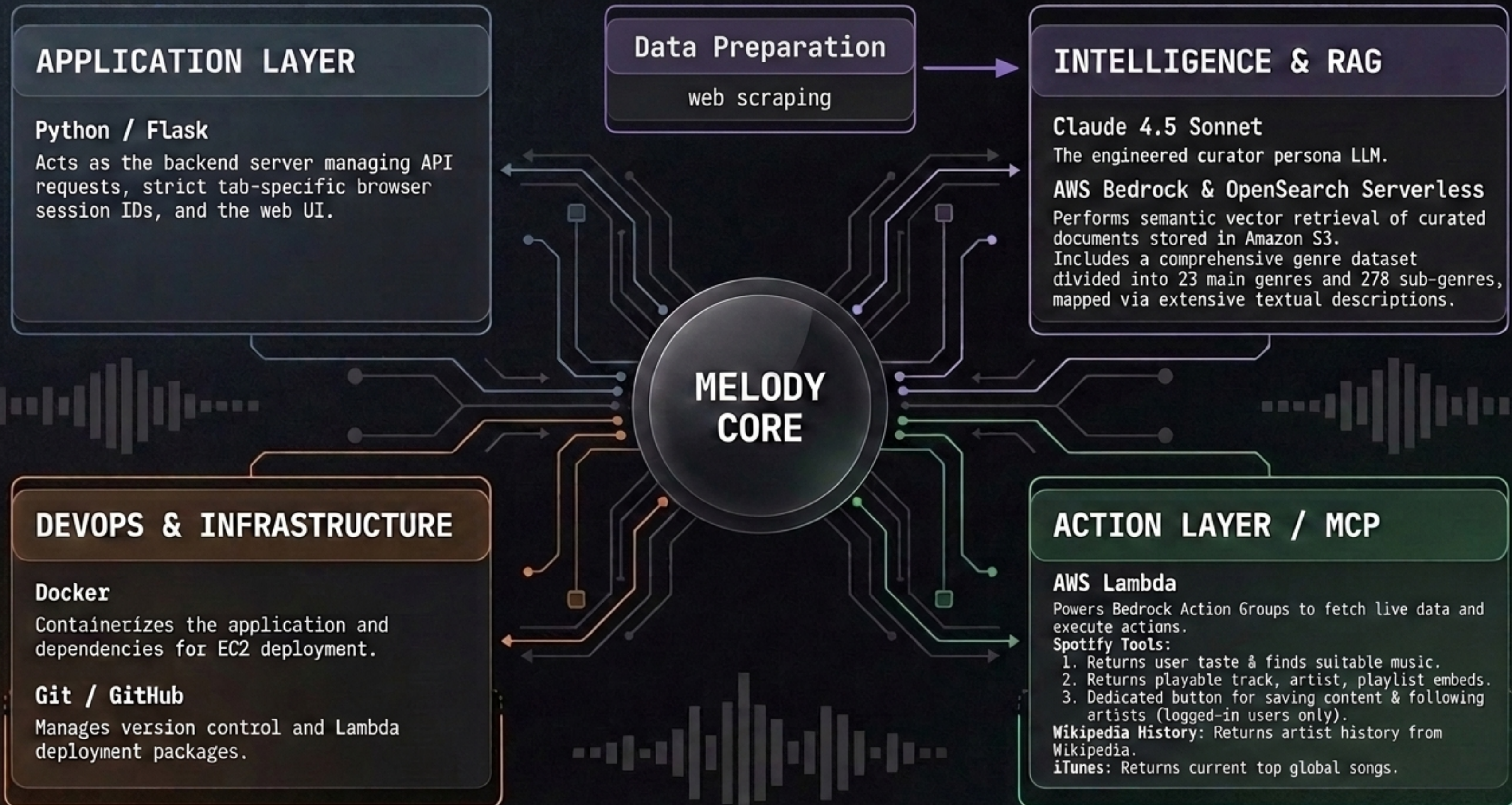
Listeners craving deep, context-aware discovery based on music history, reviews, and abstract vibes (e.g., warm, melancholy, high-energy).

THE ENGINE

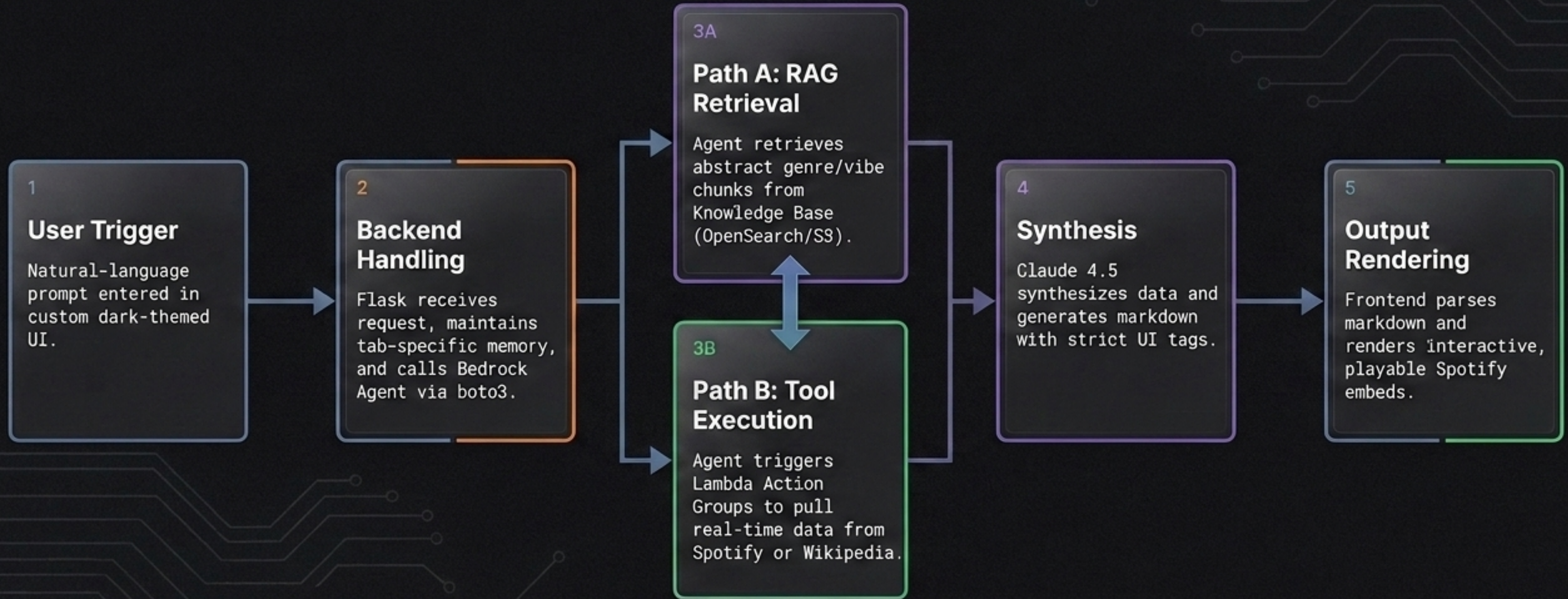
An AI assistant utilizing a custom RAG pipeline to cross-reference curated genre data with live Spotify and Wikipedia tools.

THE VALUE

Encourages listeners to step outside algorithmic comfort zones to experience music in an exploratory, emotional way.



Orchestration Flowchart



Live Demo Roadmap

01

Auth & State

Demonstrating the login screen and Spotify OAuth authentication to show real-time application state change.

02

RAG + Tool Reasoning

Executing a conceptual prompt (Songs like Bon Iver but warmer) to showcase curator reasoning, return a playable track embed, and demonstrate 'like'/'dislike' feedback

03

The Anti-Algorithm

Triggering the dedicated button designed to intentionally break established taste patterns and fetch unexpected recommendations.

04

Secondary MCP Tools

Asking for artist history (e.g., Coldplay) to trigger the Wikipedia Lambda tool, generating a chronological historical timeline without hallucination.

Retrospective

Component	Friction	Momentum
Session Orchestration	Blending abstract RAG data (vibes) with precise API data (Spotify) across multiple browser tabs.	Successfully maintained strict tab-separated session memory without data bleed.
LLM Safety & Compliance	Anthropic's strict copyright guardrails caused Claude 4.5 to rigidly refuse song lyric fetch requests.	Executed an agile pivot to implement Global Charts and Wikipedia Lambda tools, preserving robust tool usage.
Expanded Tooling Layer	Current context is limited to Spotify's ecosystem and Wikipedia.	Future iterations will scale Lambda Action Groups to include active POST operations (e.g., saving playlists directly to libraries).